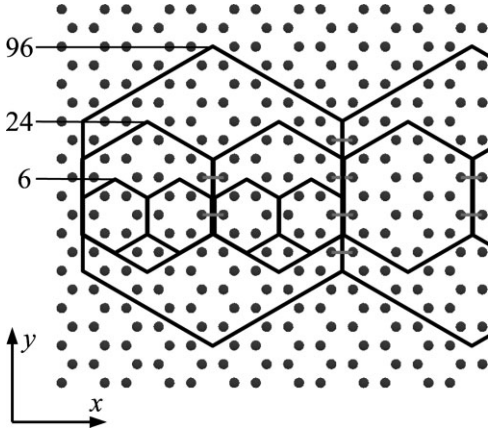
Three Routes to the Friction Matrix and Their Application to the Coarse-Graining of Atomic Lattices

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The calculation of the friction matrix in the coarse-grained (CG) description of an atomistic system is a crucial issue, in order to properly account for the dissipative effects inherent to any reduced representation of the atomistic dynamics. Within the Mori-Zwanzig projection operator approach to CG, there are several possibilities for the definition of the friction matrix, depending on the projector that is being used. In this paper, the connection of two of these projectors (Mori's and Zwanzig's) is discussed and the corresponding merits and limitations are analysed. Moreover, three different ways of computing the friction matrix in the Mori's framework are presented, along with a discussion of their mutual connections. By the example of CG centre of mass blob variables in the graphene lattice, it is shown that, even though the three approaches are equivalent from a theoretical viewpoint, they may differ considerably in terms of practical implementation for computer simulation purposes. In the given example, which is representative for atomic lattices, it turns out that a linear regression of the velocity–velocity correlation function, inspired by the Einstein–Helfand approach, is the least error-prone against disturbances from optical modes.



Introduction

The search for simplified descriptions of atomic systems, achieving an optimal trade-off between physical realism and computational viability, is a very active research area in material science and statistical physics in general.[1] Indeed, despite the impressive advances in computer technology, it is well known that the direct numerical simulation of complex systems, namely a one-to-one correspondence between the physical system itself and its simulated counterpart, remains beyond reach of even the most powerful computers within any foreseeable future. Given the limitations of full-scale direct numerical simulation, minimal models must be developed, to achieve the aforementioned tradeoff between physical realism and

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computational tractability. To this purpose, a broad variety of methods have been developed, which can generically be referred to as multiscale techniques.[2–7] Although they may differ in a series of crucial details, all multiscale techniques must appeal to some form of coarse-graining (CG), that is, a substantial reduction of the number of degrees of freedom to be retained in the computational model. Loosely speaking, the inspiring guideline for coarse-graining is deceivingly simple: 'retain the relevant degrees of freedom (DOF) and discard the irrelevant ones'. In practice, however, drawing this line turns out to be an extremely difficult task, which is usually performed through a clever mix of mathematical simplifications and heuristic physical intuition. Turning the 'art' of coarse-graining into a systematic mathematical procedure represents one of the major goals of computational statistical mechanics. Even though achieving such goal in full generality might prove a chimera, the importance of increasing the systematicity of CG procedures can be hardly overstated.

For the case of atomistic systems, the main task of coarse-graining is to link the molecular specificity of the system with its macroscopic or mesoscopic description. A well-defined theory of CG, which goes under the name of Mori–Zwanzig theory,[8–10] is available since long. The basic idea is to select a number of variables, those with the longest time scales of evolution, and derive closed equations for their dynamics. The procedure is based on an exact theorem that allows one to express Hamilton's equation (or the Liouville equation) in terms of the selected CG variables.

Under the Markovian assumption that the CG variables evolve on a time scale well separated from their time derivatives, the CG dynamic equation becomes a closed set of stochastic differential equations (SDE) for the CG variables. The effect of the eliminated degrees of freedom shows up in the SDE of the CG variables in the form of effective potentials (or equations of state), dissipation (constitutive relations) and stochastic forces. The dissipative terms are basically determined by the friction matrix, which contains the transport coefficients of the system at the specific level of description considered. The stochastic terms are also given by the friction matrix or, more precisely, by its square root, in accordance with the fluctuation–dissipation relation. The theorem alluded to above, makes use of a projection operator and, depending on the specific form of the projector, two 'flavors' of the theory, Mori or Zwanzig, are obtained. Other projectors, like Robertson's, have also been proposed,[8,11] which lead to closed equations for probability distributions.[12] The Mori projector leads to a linear SDE for the CG variables, while the Zwanzig projector leads to a non-linear SDE. In the present paper, we show that the Mori SDE can be obtained from the Zwanzig's one, under the approximation that the system is close to the equilibrium state, so that the fluctuations of the CG variables are small in some suitable sense. While the formal connection between Robertson's projector and Mori's projector is well understood,[8] we are not aware of any explicit discussion of the connection between Mori and Zwanzig projectors.

As mentioned before, Zwanzig's SDE is more fundamental than Mori's one, as it captures non-linear effects and, consequently, it can deal with situations far from equilibrium. In fact, Zwanzig's theory naturally incorporates the notion of effective potentials, defined in the usual way, in terms of conditional expectations,[13] that is, through the integration of the exponential of the Hamiltonian over the discarded degrees of freedom.

In Mori theory, only the quadratic 'small oscillations' approximation, is obtained for such potentials. A second difference between Mori's and Zwanzig's SDE concerns the form of the dissipation terms. In both cases, a friction matrix containing the transport coefficients of that level of description appears. In the Zwanzig SDE, this matrix depends, in general, on the whole set of CG variables because it is defined through a *conditional* average, where the condition is on the fact that the CG variables take specific values. In the Mori SDE, the friction matrix is defined through an *unconditional* i.e., ordinary equilibrium average, and the friction matrix is independent of the CG variables. The fact that the conditional averages that appear in the definition of the effective potential and friction matrix lead to state dependent objects represents a serious practical drawback of the Zwanzig theory. Indeed, if the number of variables is large (larger than, say, three or four), then, sampling the CG state space in order to obtain the functional form of the effective potential and friction, becomes an unfeasible task. This is the infamous *curse of dimensionality* pervading most of many-body physics.

A possible way out is to formulate approximate parameterized forms of the effective potentials and frictions, and devise methods for optimal parametric estimation. Shell's method,[14] is a powerful and elegant approach for the parameter estimation of the effective potential[14,15] and it has been recently generalized to dynamic situations.[16] Concerning the friction matrix in Zwanzig's theory, in the cases where the fluctuations of the CG variables are sufficiently small, and the equilibrium distribution function of CG variables is highly peaked, we are allowed to approximate its expression with Mori's expression for the friction matrix.

In the present paper, we focus on Mori's theory for the formulation of the friction matrix and show that three mathematically equivalent but different forms of the friction terms can be formulated. These three forms can be associated with the names of Green-Kubo, Onsager and Einstein–Helfand. While the connection between these different forms may be known, here we explore how they

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perform in terms of practical computer simulation of atomistic systems. To this end, we apply the three forms to the case of a blob-based CG procedure, as applied to a simple microscopic model of graphene.[15]

After a short summary of Mori–Zwanzig theory, where we work out the connection between Mori's and Zwanzig's SDE, we use Mori's theory to derive the Green-Kubo, Onsager and Einstein–Helfand routes to compute the friction matrix. Subsequently, we show that, for our specific example of coarse-grained (CG) graphene blobs, a combination of the Onsager and Einstein–Helfand routes is the best choice.

Mori–Zwanzig Theory in a Nutshell

At the more microscopic level of description we assume that the system is well described by classical mechanics and that the microscopic state z is given by the set of positions and momenta of all the atoms in the system. Hamilton's equations can be written in a very compact form as

$$\dot{z}(t) = Lz \tag{1}$$

where L is the Liouville operator. This set of first order differential equations has as a formal solution

$$z(t) = \exp\{Lt\}z_0 \tag{2}$$

where z_0 is the initial condition. At a macroscopic level, we have the system described by a smaller set of phase functions $A_\mu(z)$ which we choose as the CG variables. This choice, whose details shall be given shortly, is the first step in the coarse-graining process. By writing $A_\mu(t) = A_\mu(z(t))$, the evolution of these functions in phase space is given by

$$\dot{A}_\mu(t) = L\,A_\mu(t) = \exp\{Lt\}LA_\mu(z_0). \tag{3}$$

where we have used $A_\mu(\exp\{Lt\}z_0) = \exp\{Lt\}A_\mu(z_0)$, and the fact that, clearly, L commutes with $\exp\{Lt\}$. The goal of coarse-graining is to replace Equation (3) by an equation of motion which depends on the microstate z only through the CG variables $A(z)$ themselves leading to a closed equation for the CG variables. The two most prominent equations are the Mori SDE and the Zwanzig SDE, which are described in the following.

Mori SDE

Mori's exact Generalized Langevin Equation is an evolution equation for the set of CG variables $A(z)$. It is, in fact, just a rewriting of Equation (3) in the following way

$$\dot{A}_\mu(t) = \Omega_{\mu\nu}\delta A_\nu(t) - \int_0^t K_{\mu\nu}(t-u)\delta A_\nu(u)\mathrm{d}u + F^+_\mu(t) \tag{4}$$

where $\delta A_\mu(t) = A_\mu(t) - \langle A_\mu\rangle$ are the deviations of the CG variables with respect to their equilibrium average. Here and in the following the Einstein repeated index summation convention is assumed. The drift term Ω, the dissipative kernel K and the projected force F^+_μ are given respectively by

$$\begin{aligned}\Omega_{\mu\nu} &= \langle\delta A_\nu\delta A_\sigma\rangle^{-1}\langle A_\sigma LA_\mu\rangle\\ K_{\mu\nu}(t) &= \langle\delta A_\nu\delta A_\sigma\rangle^{-1}\langle F^+_\sigma(0)F^+_\mu(t)\rangle\\ F^+_\mu(t) &= \exp\{\mathcal{Q}Lt\}\mathcal{Q}LA_\mu\end{aligned} \tag{5}$$

where $\langle\cdots\rangle$ denotes an equilibrium average

$$\langle\cdots\rangle \equiv \int \mathrm{d}z\rho^{\mathrm{eq}}(z)\cdots \tag{6}$$

and $\rho^{\mathrm{eq}}(z)$ is the equilibrium ensemble. The projection operator $\mathcal{Q}$ is defined as $\mathcal{Q} = 1-\mathcal{P}$ where $\mathcal{P}$ is also a projector, whose effect on an arbitrary phase function $F(z)$ is given by

$$\mathcal{P}F(z) = \langle F\rangle + \delta A_\nu(z)\langle\delta A_\nu\delta A_\mu\rangle^{-1}\langle\delta A_\mu F\rangle. \tag{7}$$

The Mori projector (7) satisfies $\mathcal{P}\delta A_\mu = \delta A_\mu$ and transforms an arbitrary function of phase space into a linear combination of the CG variables. The projected forces have zero mean $\langle F^+_\mu(t)\rangle = 0$ and are uncorrelated from previous values of the CG variables $\langle\delta A_\sigma F^+_\mu(t)\rangle = 0$ for $t \geq 0$. The Markovian approximation is taken under the assumption that the coarse variables evolve in a time scale much larger than the memory time scale. In this case, one may approximate

$$K_{\mu\nu}(t) \approx \Gamma_{\mu\nu}\delta^+(t) \tag{8}$$

where $\Gamma_{\mu\nu}$ is called the friction matrix and it is given by

$$\Gamma_{\mu\nu} \equiv \langle\delta A_\nu\delta A_\sigma\rangle^{-1}\int_0^\tau \mathrm{d}t\langle F^+_\sigma(0)F^+_\mu(t)\rangle. \tag{9}$$

Here τ is a time in which the memory has decayed (i.e., the 'support' of the delta function $\delta^+(t)$). Equation (9) is the so-called Green–Kubo expression for the friction matrix. Note that

$$F^+_\mu(t) = \frac{\mathrm{d}}{\mathrm{d}t}\exp\{\mathcal{Q}Lt\}A_\mu \tag{10}$$

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and, therefore

$$\int_0^\tau dt\langle F_\mu^+(0)F_\nu^+(t)\rangle = \langle F_\mu^+(0)\exp\{\mathcal{Q}L\tau\}A_\mu\rangle - \langle F_\mu^+(0)A_\mu(0)\rangle = \langle F_\mu^+(0)\exp\{\mathcal{Q}L\tau\}A_\mu\rangle \quad (11)$$

For $\tau\to\infty$ we expect a decorrelation in the form $\langle F_\mu^+(0)\exp\{\mathcal{Q}L\tau\}A_\mu\rangle \to \langle F_\mu^+\rangle\langle A_\mu\rangle = 0$. Therefore, the friction matrix suffers from the plateau problem.[17,18]

Under the Markovian approximation Equation (8), the exact Equation (4) becomes

$$\dot{A}_\mu(t) = \Omega_{\mu\nu}\delta A_\nu(t) - \Gamma_{\mu\nu}\delta A_\nu(t) + F_\mu^+(t) \quad (12)$$

While $\Omega_{\mu\nu}$ could be computed from time averages in a molecular dynamics (MD) simulation that samples the equilibrium ensemble, the friction matrix cannot be computed directly from such a simulation, because it involves the projected dynamics $\exp\{\mathcal{Q}Lt\}$, instead of the real dynamics $\exp\{Lt\}$ which is the one obtained from MD. Below, we will present three different methods that may be used for such a calculation, and discuss their mutual consistency.

Zwanzig SDE

Zwanzig theory makes use of the projector

$$\mathrm{P}F(z) = \langle F\rangle^{\alpha=A(z)} \quad (13)$$

where the conditional average is defined by

$$\langle\cdots\rangle^\alpha = \frac{1}{P^{\mathrm{eq}}(\alpha)}\int dz\rho^{\mathrm{eq}}(z)\delta(A(z)-\alpha)\cdots, \qquad P^{\mathrm{eq}}(\alpha) = \int dz\rho^{\mathrm{eq}}(z)\delta(A(z)-\alpha). \quad (14)$$

Here $P^{\mathrm{eq}}(\alpha)$ (not to be confused with the projector P) is the measure of the submanifold $A(z)=\alpha$ and it is also the equilibrium distribution of the CG variables. The entropy (at the level of description of α) is the logarithm of 'the number of microstates compatible with the macrostate α', i.e. $S(\alpha) = k_B\ln P^{\mathrm{eq}}(\alpha)$. The Zwanzig projector (13) transforms any function of the microscopic state into a function that depends on the microscopic state through the CG variables.

The evolution equation for the phase functions $A_\mu(t)$ is[8,19–21]

$$\dot{A}_\mu(t) = \langle LA_\mu\rangle^{A(t)} + M_{\mu\nu}(A(t))\frac{\partial S}{\partial\alpha_\nu}(A(t)) + k_B\frac{\partial M_{\mu\nu}}{\partial\alpha_\nu}(A(t)) + \tilde{R}_\mu(t), \quad (15)$$

where the friction matrix is expressed by its Green–Kubo form

$$M_{\mu\nu}(\alpha) = \frac{1}{k_B}\int_0^\tau dt'\langle\tilde{R}_\mu(0)\tilde{R}_\nu(t')\rangle^\alpha, \quad (16)$$

and it is a symmetric and positive definite matrix. The projected current, also called the random force, is

$$\tilde{R}_\mu(t) = \exp\{tQL\}QLA_\mu. \quad (17)$$

The only approximation used to derive Equation (15) is, again, that the correlation time of $\langle\tilde{R}_\mu\tilde{R}_\nu(t)\rangle^\alpha$ is much smaller than the time scale of variation of $A(t)$.

Mori's SDE is an Approximation of Zwanzig's SDE

We now show under what approximations the Mori SDE (12) can be understood as a particular case of the Zwanzig SDE (15). To our knowledge, the formal connection of the Mori and Zwanzig SDEs has not been discussed explicitly in the past. The connection of Mori's with Robertson's projector is discussed in ref.[8] To this end we elaborate a bit the reversible term $\langle LA_\mu\rangle^\alpha$. By integrating by parts the Liouville operator, we may write

$$\begin{aligned}\langle LA_\mu\rangle^\alpha &= -\frac{1}{P^{\mathrm{eq}}(\alpha)}\int dz\rho^{\mathrm{eq}}(z)A_\mu(z)L\delta(A(z)-\alpha)\\ &= \frac{1}{P^{\mathrm{eq}}(\alpha)}\frac{\partial}{\partial\alpha_\nu}\int dz\rho^{\mathrm{eq}}(z)\delta(A(z)-\alpha)A_\mu(z)LA_\nu(z)\\ &= \frac{1}{P^{\mathrm{eq}}(\alpha)}\frac{\partial}{\partial\alpha_\nu}P^{\mathrm{eq}}(\alpha)\langle A_\mu LA_\nu\rangle^\alpha\\ &= \frac{\partial}{\partial\alpha_\nu}\langle A_\mu LA_\nu\rangle^\alpha + \frac{1}{k_B}\langle A_\mu LA_\nu\rangle^\alpha\frac{\partial S}{\partial\alpha_\nu}(\alpha)\end{aligned} \quad (18)$$

This is an exact result. We can now make the assumption that the reversible matrix $\langle A_\mu LA_\nu\rangle^\alpha$ can be approximated by its equilibrium average, this is

$$\langle A_\mu LA_\nu\rangle^\alpha \approx \int d\alpha P^{\mathrm{eq}}(\alpha)\langle A_\mu LA_\nu\rangle^\alpha = \langle A_\mu LA_\nu\rangle. \quad (19)$$

We expect this approximation to be correct whenever the system is near equilibrium and thermal fluctuations, i.e., the range of variation of the CG variables, is small. In that case $P^{\mathrm{eq}}(\alpha)$ is highly peaked. The matrix, therefore, becomes independent of the CG variables and Equation (18) becomes

$$\langle LA_\mu\rangle^\alpha \approx \frac{1}{k_B}\langle A_\mu LA_\nu\rangle\frac{\partial S}{\partial\alpha_\nu}(\alpha) \quad (20)$$

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and the whole state dependence of the reversible part is through the entropy function. If we now assume that the highly peaked equilibrium probability distribution of the CG variables $P^{eq}(\alpha)$ can be approximated by a Gaussian, then the corresponding entropy function becomes the following quadratic function of the CG variables (up to irrelevant constant terms)

$$S(\alpha) = -\frac{k_B}{2}\delta\alpha_\mu \sigma_{\mu\nu}\delta\alpha_\nu + \text{const.} \tag{21}$$

where σ is the inverse of the matrix of covariance,

$$\langle \delta A_\mu \delta A_\nu \rangle = \sigma^{-1}_{\mu\nu} \tag{22}$$

and $\delta\alpha_\mu = \alpha_\mu - \langle A_\mu \rangle$. With this Gaussian approximation, the reversible term finally becomes

$$\langle LA_\mu \rangle^\alpha \approx -\langle A_\mu LA_\nu \rangle \langle \delta A_\nu \delta A_\sigma \rangle^{-1} \delta\alpha_\sigma \tag{23}$$

which is identical to the first term $\Omega_{\mu\nu}\delta A_\nu(t)$ in Equation (12), because $-\langle A_\mu LA_\nu \rangle = \langle A_\nu LA_\mu \rangle$ as can be shown by integration by parts of the Liouville operator.

Let us now turn to the dissipative part of the Zwanzig SDE (15) and assume in a similar way that the friction matrix defined in Equation (16) can be approximated by its equilibrium average

$$\begin{aligned} M_{\mu\nu}(\alpha) &\approx M^{eq}_{\mu\nu} \equiv \int d\alpha P^{eq}(\alpha) M(\alpha) \\ &= \int_0^\tau dt \langle \tilde{R}_\mu(0)\tilde{R}_\nu(t) \rangle \end{aligned} \tag{24}$$

Of course, we still have to relate the projected force $\tilde{R}(t)$ with the projected force $F^+_\mu(t)$ and this requires the consideration of the relationship between the Mori and Zwanzig projectors. The connection is as follows. The conditional average of an arbitrary phase function $\langle F \rangle^\alpha$ is a function of the CG variables α. If the range of values that take these variables is small, this is, we are near equilibrium in a situation in which the fluctuations are small (away from critical points, for example), we may approximate the functional form of the conditional average with a linear function of the form

$$\langle F \rangle^\alpha = a + b_\mu \delta\alpha_\mu \tag{25}$$

The values of the constants a, b_μ are obtained by requiring the fulfilment of the following two exact relationships

$$\begin{aligned} \int d\alpha P^{eq}(\alpha) \langle F \rangle^\alpha &= \langle F \rangle \\ \int d\alpha P^{eq}(\alpha) \delta\alpha_\mu \langle F \rangle^\alpha &= \langle \delta A_\mu F \rangle \end{aligned} \tag{26}$$

These identities can be checked by inserting the definition of the conditional average (14) and performing the integral over the delta functions. The final result for the linear approximation is

$$\langle F \rangle^\alpha = \langle F \rangle + \langle F \delta A_\mu \rangle \langle \delta A_\mu \delta A_\nu \rangle^{-1} \delta\alpha_\mu \tag{27}$$

This equation shows that in the near equilibrium limit the Zwanzig projector (13) and the Mori projector (7) are identical. Therefore, in the near equilibrium, small fluctuations regime, the Zwanzig friction matrix may be approximated by the Mori friction matrix. In addition, if we assume that the entropy is a quadratic function then we will end up with a dissipative contribution in the Zwanzig SDE that is exactly given by the dissipative part of the Mori SDE (12).

Three Routes to the Friction Matrix Γ

In this section, we present three different expressions for the Mori friction matrix that go under the names of Green-Kubo, Onsager and Einstein-Helfand and show the way they are connected.

Green–Kubo

The first method to compute the friction matrix is from the Green–Kubo expression (9). The usual approximation is to assume that, for short times, we can neglect the time integral in Equation (4), and we have an approximation of the projected forces F_μ as

$$\begin{aligned} F^+_\mu(t) &\approx \frac{d}{dt}A_\mu(t) - \Omega_{\mu\nu}A_\nu(t) \\ &= \exp\{Lt\}\mathcal{Q}LA_\mu \end{aligned} \tag{28}$$

where we have used $\mathcal{Q}LA_\mu = LA_\mu - \Omega_{\mu\nu}A_\nu$ as can be checked from the definition of the projector (7). Note that this approximation is equivalent to substitute the projected dynamics $\exp\{\mathcal{Q}Lt\}A$ with the real dynamics $\exp\{Lt\}$ in the definition of the projected force in Equation (5).

By inserting this approximation in the Green–Kubo expression (9) we obtain

$$\Gamma_{\mu\nu} = \langle A_\nu A_\sigma \rangle^{-1} \int_0^\tau \langle \mathcal{Q}LA_\sigma \exp\{Lt\}\mathcal{Q}LA_\mu \rangle \, dt \tag{29}$$

As opposed to (9), the approximated form for friction matrix (29) becomes computable from a standard MD simulation because it involves the real dynamics $\exp\{Lt\}$, not the projected dynamics $\exp\{\mathcal{Q}Lt\}$. It looks as if this Green–Kubo calculation is 'microscopic' because it uses the

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correlation of the 'forces' LA_μ computed during a short time interval τ.

Onsager

The second method for the calculation of the friction matrix relies on Onsager's regression hypothesis.[22] This hypothesis can be *deduced* in the present formulation as follows. By multiplying Equation (12) with $A(z)$ and averaging over the initial conditions with respect to the equilibrium ensemble we obtain an equation for the equilibrium correlations

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \delta A_\mu(t)\delta A_\nu\rangle = (\Omega_{\mu\sigma}-\Gamma_{\mu\sigma})\langle \delta A_\sigma(t)\delta A_\nu\rangle. \tag{30}$$

This Equation (30) can be interpreted as Onsager's regression of fluctuations that states that 'fluctuations' (or rather their correlations) evolve in the same way as the averages of the macroscopic variables [i.e., the result of averaging of Equation (12)]. Note that within Mori theory the matrix of correlations has always an exponential decay (that due to $\Omega_{\mu\nu}$ may have oscillatory components). All the objects in Equation (30) except the friction matrix $\Gamma_{\mu\nu}$, can be explicitly computed in a MD simulation, because the real dynamics is involved in the correlation functions. Of course, the time scale of the relevant variables $A_\mu(t)$ has been assumed to be large and therefore, to simulate the macroscopic Equation (30) for very long times is very costly. However, the equation should be, in principle valid for all times, so we could particularize the above equation for $t=0$. If we let $t\to 0$ in Equation (30) we have a method to obtain the friction out of the macroscopic evolution of correlations, and static quantities.

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \delta A_\mu(t)\delta A_\nu\rangle\Big|_{t=0} = (\Omega_{\mu\sigma}-\Gamma_{\mu\sigma})\langle \delta A_\sigma\delta A_\nu\rangle \tag{31}$$

We may solve for $\Gamma_{\mu\nu}$

$$\Gamma_{\mu\nu} = -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\left[\frac{\mathrm{d}}{\mathrm{d}t}\langle \delta A_\sigma\delta A_\mu(t)\rangle\Big|_{t=0} - \langle \delta A_\sigma\delta LA_\mu\rangle\right] \tag{32}$$

We call this the Onsager's route to compute the friction matrix. As opposed to the Green–Kubo route, it looks like this route is 'macroscopic' because it uses the time evolution of the macroscopic variables themselves.

The first thing to note is that Equation (32) *is inconsistent*, because

$$\frac{\mathrm{d}}{\mathrm{d}t}A_\mu(t)\Big|_{t=0} = LA_\mu \tag{33}$$

and then the right hand side of Equation (32) vanishes, leading to a zero friction matrix! The origin of this inconsistency can be easily traced back to the fact that Equation (12), from which Equation (32) follows directly, is only valid when t is larger than the correlation time of the memory function, and cannot be used strictly at time $t=0$. What we expect is that the rhs of Equation (32) will be zero at $t=0$, it will increase rapidly, within the timescale given by the memory time scale, and afterwards it will start to decay slowly, with the time scale of the relevant variables $A(z)$. The maximum of this curve is expected to be around τ, the correlation time of the memory. Onsager's route should then be properly written as

$$\Gamma_{\mu\nu} = -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\left[\frac{\mathrm{d}}{\mathrm{d}t}\langle \delta A_\sigma\delta A_\mu(t)\rangle\Big|_{t=\tau} - \langle \delta A_\sigma L\delta A_\mu\rangle\right] \tag{34}$$

The consistency between the 'microscopic' Green–Kubo expression (29) and the 'macroscopic' Onsager expression (34) for the friction matrix can now be easily assessed. Let us start from Equation (29)

$$\begin{aligned}
\Gamma_{\mu\nu} &= \langle \delta A_\nu\delta A_\sigma\rangle^{-1}\int_0^\tau \langle QLA_\sigma \exp\{Lt\}QLA_\mu\rangle \mathrm{d}t \\
&= \langle \delta A_\nu\delta A_\sigma\rangle^{-1}\int_0^\tau \langle LA_\sigma \exp\{Lt\}QLA_\mu\rangle \mathrm{d}t \\
&= -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\int_0^\tau \frac{\mathrm{d}}{\mathrm{d}t}\langle A_\sigma \exp\{Lt\}QLA_\mu\rangle \mathrm{d}t \\
&= -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\langle A_\sigma \exp\{L\tau\}QLA_\mu\rangle \\
&= -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\langle A_\sigma \exp\{L\tau\}(LA_\mu-\Omega_{ii'}A_{i'}\rangle \\
&= -\langle \delta A_\nu\delta A_\sigma\rangle^{-1}\frac{\mathrm{d}}{\mathrm{d}t}\langle \delta A_\sigma\delta A_\mu(t)\rangle\Big|_{t=\tau} \\
&\quad + \langle \delta A_\nu\delta A_\sigma\rangle^{-1}\Omega_{ii'}\langle \delta A_\sigma\delta A_{i'}(\tau)\rangle
\end{aligned} \tag{35}$$

For τ small, we expect that $\langle \delta A_\sigma\delta A_{i'}(\tau)\rangle \approx \langle \delta A_\sigma\delta A_{i'}\rangle$ and, therefore $\Omega_{ii'}\langle \delta A_\sigma\delta A_{i'}(\tau)\rangle \approx \langle A_\sigma LA_\mu\rangle$. We then recover the Onsager's route of the friction matrix, Equation (34), from the Green–Kubo's route.

Einstein–Helfand

We can obtain yet another route for the calculation of the friction matrix. Consider the 'rms displacement' of the fluctuations of the CG variables

$$\begin{aligned}
&\frac{1}{2}\langle(\delta A_\mu(t)-\delta A_\mu(0))(\delta A_\nu(t)-\delta A_\nu(0))\rangle \\
&= \langle \delta A_\mu\delta A_\nu\rangle - \langle \delta A_\mu\delta A_\nu(t)\rangle
\end{aligned} \tag{36}$$

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where we have made use of the stationarity properties of the equilibrium correlations. Next write

$$\langle \delta A_\mu \delta A_\nu \rangle - \langle \delta A_\mu \delta A_\nu(t) \rangle = -\int_0^t \frac{\mathrm{d}}{\mathrm{d}t'} \langle \delta A_\mu \delta A_\nu(t') \rangle = -(\Omega_{\mu\sigma} - \Gamma_{\mu\sigma}) \int_0^t \mathrm{d}t' \langle \delta A_\sigma \delta A_\nu(t') \rangle \tag{37}$$

where in the last equation we have used Equation (30). Now, if we consider t large enough for Equation (30) to be valid, but small enough for the correlation of relevant variables to have changed appreciably so that we may approximate $\langle \delta A_\mu \delta A_\nu(t) \rangle \approx \langle \delta A_\mu \delta A_\nu \rangle$ within the integral in Equation (37), we obtain from (37)

$$\frac{1}{2} \langle (\delta A_\mu(t) - \delta A_\mu(0))(\delta A_\nu(t) - \delta A_\nu(0)) \rangle \approx -(\Omega_{\mu\sigma} - \Gamma_{\mu\sigma}) \langle \delta A_\sigma \delta A_\nu \rangle t \tag{38}$$

By computing the 'rms displacement' of the relevant variables, this equation allows to obtain the friction matrix Γ. Equation (38) is the Einstein–Helfand form for the matrix of transport coefficients[23] that tells that the rms displacement grows linearly in time. Note that this linear growth in time cannot extend indefinitely. From Equation (30) we observe that the correlation matrix decays exponentially (in matrix sense). Therefore, from Equation (36), we observe that the linear growth is limited to the time scale in which the exponential can be approximated by a linear function, i.e. for time scales barely larger than τ.

Example: Calculation of the Friction Matrix for Graphene Blobs

In ref.[15] we have presented the coarse-graining of a simple all-atom 2D microscopic model of the graphene lattice, in terms of 'blobs' described by centre of mass (CoM) variables. The equations of motion of the CG variables take the form of dissipative particle dynamics (DPD).[24,25] The CG conservative forces and the friction of the DPD model have been obtained via a *bottom-up* procedure from MD simulations as opposed to previous heuristic top-down approaches.[26,27] The separation of timescales for blobs of 24 and 96 carbon atoms, turned out to be sufficiently pronounced for the Markovian assumption, inherent to the DPD model, to provide satisfactory results. In particular, the MD velocity auto-correlation function of the blobs is well reproduced by the DPD model, provided that the effect of friction and noise is taken into account. However, DPD cross-correlations between neighbouring blobs show some appreciable discrepancies with respect to the MD results.[15]

In the following subsections, we will first briefly summarize this DPD-model. Then we present in detail the evaluation of the friction matrix, by following the three routes described above and assess what are the advantages, of one over the others.

Blob Dynamics in the Honeycomb Lattice

Graphene is a 2D arrangement of carbon atoms in a honeycomb lattice structure. Figure 1 shows the two-dimensional honeycomb lattice of the C-atoms where each C-atom is strongly bound to three neighbours. We consider here the in-plane 2D motion of the system. The dynamics of the graphene model is described by the Hamiltonian equations of motion for the set of microscopic variables $\{\mathbf{r}_i, \mathbf{p}_i\}$, where m_i is the mass, $\mathbf{r}_i$ the position and $\mathbf{p}_i$ the momentum of C-atom i. For the sake of simplicity, a simple two dimensional force field based on pairwise and angular springs is used. The techniques presented in the following sections to determine the friction matrix are independent of the specific Hamiltonian that is used to represent the microscopic model. Specifically, the force $\mathbf{F}_i$ on the i-th C-atom is defined by

$$\mathbf{F}_i = \sum_{P_{ij}} \mathbf{F}_{ij} + \sum_{T_{ijk}} \mathbf{F}_{ijk} \tag{39}$$

where P_{ij} denotes all pairs of C-atoms i and j which are bonded in the graphene sheet and T_{ijk} denotes all triplets of particles $\{i, j, k\}$. The pair force is defined as

$$\mathbf{F}_{ij}(r_{ij}) = -k_s(r_{ij} - r_0)\mathbf{e}_{ij}, \tag{40}$$

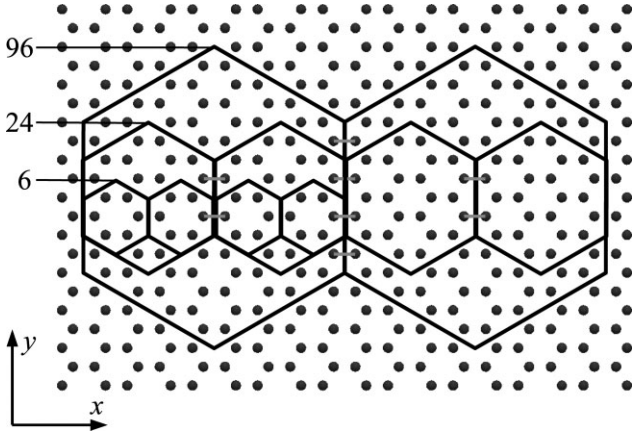


Figure 1. Construction of the different blobs and their arrangement in the graphene lattice. The integer numbers indicate the number of C-atoms in one blob, i.e., the coarse-graining number N_{cg}.

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where $k_s > 0$ is a stiffness constant, r_0 the equilibrium bond-length, $r_{ij} = |\mathbf{r}_{ij}|$, $\mathbf{e}_{ij} = \mathbf{r}_{ij}/r_{ij}$ and $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. The angular force is $\mathbf{F}_{ijk}(\alpha) = -\nabla_{\mathbf{r}_i} U(\alpha_{ijk})$ where the potential energy is defined as

$$U(\alpha_{ijk}) = \frac{1}{2} k_a \left(\cos\alpha_{ijk} - \cos\alpha_0\right)^2 \tag{41}$$

with the equilibrium angle $\alpha_0 = 2\pi/3$ and

$$\cos\left(\pi - \alpha_{ijk}\right) = \frac{\mathbf{r}_{ij}\mathbf{r}_{jk}}{r_{ij}r_{jk}}. \tag{42}$$

Here k_a is a second stiffness constant for angular changes.

The unit of length l^* is fixed by setting $l^* \equiv r_0 \approx 0.142$ nm in graphene. The unit of mass is $m^* \equiv m_C$ where $m_C \approx 2 \times 10^{-26}$ kg is the mass of one C-atom. We choose the unit of time $t^* \equiv (m_C/k_s)^{1/2}$, which amounts to set the value of $k_s = 1$ in the selected units. In addition, we set the value of the angular spring constant in our model to $k_a = 1$ in the units selected. We estimate k_s by comparing the speed of sound measured in our MD simulation with that of actual graphene.[28] For our selected unit of time, this yields $t^* \approx 9.56$ fs. The unit of temperature is $T^* \equiv m_C r_0^2/k_B t^{*2} \sim 3.2 \times 10^5$ K. MD-simulations are performed in the (N,V,E)-ensemble at a temperature of $T = 0.002$ (corresponding to $T \approx 640\,K$).

In the present work, we describe graphene at a CG level in terms of lumps of carbon atoms, described through their centre of mass position and momentum variables. Figure 1 shows how the blobs of different sizes are constructed.[26] For the sake of clarity, just those interatomic bonds are shown, which connect different blobs to each other. The figure shows three different blob sizes corresponding to three different levels of coarse-graining. Two such levels, $N_{cg} = 24$, 96, will be analyzed in this work, where N_{cg} is the coarse-graining number, i.e., the number of C-atoms contained in one blob. As one can easily see from Figure 1, the intermediate level $N_{cg} = 54$ and of course higher coarse-graining numbers are possible as well.[15]

Due to the hexagonal structure of the blobs each blob has six neighbours of the same size. For special sizes of the periodic graphene sheet the periodicity is preserved for each investigated blob size. In particular, two periodic graphene sheets are used in this work, with a number of C-atoms of $N_C = 10\,368$ and $N_C = 93\,312$ giving, respectively, 432 and 3888 blobs with $N_{cg} = 24$, and 108 and 972 blobs with $N_{cg} = 96$.

The coarse-graining approach considered here is based on the CoM variables $\{\mathbf{R}_\mu, \mathbf{P}_\mu\}$ of blobs. Following the indexing of, e.g., ref.,[29] each blob is uniquely identified by a Greek index μ. Since the blobs are collections of C-atoms, we can define the CoM variables as follows:

$$\begin{aligned} \mathbf{R}_\mu &= \frac{1}{M_\mu} \sum_i m_i \mathbf{r}_i \delta_\mu(i) \\ \mathbf{P}_\mu &= \sum_i \mathbf{p}_i \delta_\mu(i) \end{aligned} \tag{43}$$

where the symbol $\delta_\mu(i)$ takes the value 1 if atom i is in blob μ and zero otherwise, and $M_\mu = \sum_i m_i \delta_\mu(i)$ is the total mass of blob μ. By using Zwanzig theory, it has been shown rigorously[19,30] that the resulting evolution equations for the CoM variables are those of the DPD model.[24,25] The DPD equations of motion for the position $\mathbf{R}_\mu$ and momentum $\mathbf{P}_\mu$ of a blob μ are

$$\begin{aligned} \dot{\mathbf{R}}_\mu(t) &= \frac{\mathbf{P}_\mu(t)}{M_\mu} \\ \dot{\mathbf{P}}_\mu(t) &= \mathbf{F}_\mu^C(t) + \mathbf{F}_\mu^D(t) + \mathbf{F}_\mu^R(t), \end{aligned} \tag{44}$$

where $\mathbf{F}_\mu^C$ is a conservative force, $\mathbf{F}_\mu^D$ a dissipative force, and $\mathbf{F}_\mu^R$ is a stochastic force, the latter two being related to each other via a fluctuation–dissipation theorem. These forces read as follows:

$$\begin{aligned} \mathbf{F}_\mu^C(t) &= -\frac{\partial V^{cg}}{\partial \mathbf{R}_\mu}(R) \\ \mathbf{F}_\mu^D(t) &= -\sum_\nu \mathbf{M}_{\mu\nu}(R) \frac{\mathbf{P}_\nu(t)}{M_\mu} \end{aligned} \tag{45}$$

where $V^{cg}(R)$ is a CG potential depending on the positions of the blobs $R = \{\mathbf{R}_\mu\}$ and $\mathbf{M}_{\mu\nu}(R) = \mathbf{M}_{\nu\mu}(R)$ is a friction tensor that satisfies $\sum_\nu \mathbf{M}_{\mu\nu}(R) = 0$, so that the dissipative force conserves momentum. Traditionally, one writes the dissipative force as $\mathbf{F}_\mu^D(t) = -\sum_\nu \mathbf{M}_{\mu\nu}(R)\mathbf{V}_{\mu\nu}$, where $\mathbf{V}_{\mu\nu}$ is the relative velocity between blobs, so that Galilean invariance is manifest. The above forces are given in terms of conditional averages that can be computed, in principle, from constrained MD simulations.[19] Equations (44) are the manifestation of Equation (15) for $A_\mu = \{\mathbf{R}_\mu, \mathbf{P}_\mu\}$. The main approximation leading to Equation (44) is the Markovian assumption (8). Then the memory kernel can be approximated by

$$< \mathbf{F}_\mu^R(t')\mathbf{F}_\nu^R(t) >^R \approx 2k_B T \mathbf{M}_{\mu\nu}(R)\delta(t'-t) \tag{46}$$

where $\mathbf{M}_{\mu\nu}(R)$ is given by the Green–Kubo formula (16). Equation (46) implies that the projected force may be modelled as a white noise, and memory effects can be disregarded. In ref.[15] this was found to be sufficiently fulfilled for $N_{cg} = 24$ and $N_{cg} = 96$ and the DPD-forces have been determined based on blob-velocity-velocity correlation functions (VVs) computed from MD.

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In the present work we focus in more detail on techniques for the derivation of the dissipative forces of the pair-wise additive form. The determination of the conservative forces has been described in ref.[15]

Dissipative and Random Forces

In ref.[15] we proposed a particular parameterized functional form for $\mathbf{M}_{\mu\nu}(R)$ and used unconstrained MD simulations to obtain the values of the parameters. The model for the friction tensor is

$$\begin{aligned}\mathbf{M}_{\mu\nu}(R) =&\delta_{\mu\nu}\sum_{\sigma}\left[\gamma_{||}\mathbf{e}_{\mu\sigma}\mathbf{e}_{\mu\sigma}+\gamma_{\perp}(\mathbf{1}-\mathbf{e}_{\mu\sigma}\mathbf{e}_{\mu\sigma})\right]\\&-\left[\gamma_{||}\mathbf{e}_{\mu\nu}\mathbf{e}_{\mu\nu}+\gamma_{\perp}(\mathbf{1}-\mathbf{e}_{\mu\nu}\mathbf{e}_{\mu\nu})\right]\end{aligned} \tag{47}$$

that fulfills the momentum conserving requirement $\sum_{\nu}\mathbf{M}_{\mu\nu}(R)=0$. Here $\gamma_{||}$, $\gamma_{\perp}$ are two friction constants and we have introduced the unit vectors $\mathbf{e}_{\mu\nu}=\mathbf{R}_{\mu\nu}/R_{\mu\nu}$, $\mathbf{R}_{\mu\nu}\equiv\mathbf{R}_{\mu}-\mathbf{R}_{\nu}$, $R_{\mu\nu}\equiv|\mathbf{R}_{\mu\nu}|$. The functional form in Equation (47) has been used previously in refs.[19,31] In addition, it is assumed that the friction is zero for those blob pairs which are not connected.

Equation (46) is to be understood as the fluctuation–dissipation theorem for the present model giving the amplitude of the random forces in terms of the transport coefficients of the system. The random forces need to be specified as linear combinations of the Wiener process as, for example, $\mathbf{F}^{R}_{\mu}\mathrm{d}t=\mathrm{d}\tilde{\mathbf{P}}_{\mu}=\sum_{\nu}\mathbf{B}_{\mu\nu}\mathrm{d}\mathbf{W}_{\nu}(t)$. The explicit form of the noise terms is given in ref.[15]

As it is apparent, the friction matrix (47) depends on the full set of CG variables $\mathbf{R}_{\mu}$. In this paper, we advocate for the Mori approximation for the friction matrix in which the conditional average is replaced by the equilibrium average and the unit vectors $\mathbf{e}_{\mu\nu}$ are those corresponding to the equilibrium positions of the lattice.

Green–Kubo and Onsager for Blobs

For the Green–Kubo approach (29) the friction matrix reads

$$\Gamma_{\mu\nu}=\frac{1}{k_BT}\int_0^{\tau_{\mathbf{F}}}\mathrm{d}t\langle\delta\mathbf{F}_{\mu}\delta\mathbf{F}_{\nu}(t)\rangle \tag{48}$$

with

$$\delta\mathbf{F}_{\mu}=L\mathbf{P}_{\mu}-k_BT\mathbf{X}_{\nu}\langle\mathbf{X}_{\nu}\mathbf{X}_{\mu}\rangle^{-1}=\mathbf{F}_{\mu}-\mathbf{F}^{C}_{\mu} \tag{49}$$

where $\mathbf{X}_{\mu}\equiv\mathbf{R}_{\mu}-\mathbf{R}^{0}_{\mu}$ and $\mathbf{R}^{0}_{\mu}$ is the equilibrium position of blob μ. The microscopic force $\mathbf{F}_{\mu}$ is defined by

$$\mathbf{F}_{\mu}=\sum_{\nu}\mathbf{F}_{\mu\nu}\equiv\sum_{\nu}\sum_{ij}\mathbf{F}_{ij}\delta_{\mu}(i)\delta_{\nu}(j) \tag{50}$$

and is therefore accessible from standard MD-simulations. The time $\tau_{\mathbf{F}}$ is the time of the initial decay of the microscopic blob force–force correlation $\langle\mathbf{F}_{\mu}\mathbf{F}_{\nu}(t)\rangle$ (FF).

It was shown[32–34] that, if the dynamics of the relevant variables is approximately Markovian and the FF has a dominant rapidly decaying part with small $\tau_{\mathbf{F}}$ and just small terms evolving on macroscopic time scales then it is possible to calculate the friction matrix through the Kirkwood expression[18]

$$\Gamma_{\mu\nu}=\frac{1}{k_BT}\int_0^{\tau_{\mathbf{F}}}\mathrm{d}t\langle\mathbf{F}_{\mu}\mathbf{F}_{\nu}(t)\rangle. \tag{51}$$

As opposed to (48), the FF appears in this integral.

For the Onsager approach (34) the friction matrix reads[15]

$$\Gamma_{\mu\nu}=-\frac{M_{\mu}M_{\nu}}{k_BT}\frac{\mathrm{d}}{\mathrm{d}t}\langle\mathbf{V}_{\mu}\mathbf{V}_{\nu}(t)\rangle\Big|_{t=\tau_{\mathbf{F}}} \tag{52}$$

or by inserting the model (47) for $\mathbf{\Gamma}_{\mu\nu}$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\langle\mathbf{V}_{\mu}\mathbf{V}_{\mu}(t)\rangle\Big|_{t\sim\tau_{\mathbf{F}}}&=-\frac{6k_BT}{M_{\mu}^{2}}\left(\gamma_{||}+\gamma_{\perp}\right)\\\frac{\mathrm{d}}{\mathrm{d}t}\langle\mathbf{V}_{\mu}\mathbf{e}^{0}_{\mu\nu}\mathbf{V}_{\nu}(t)\mathbf{e}^{0}_{\mu\nu}\rangle\Big|_{t\sim\tau_{\mathbf{F}}}&=\frac{k_BT}{M_{\mu}M\nu}\gamma_{||}\end{aligned} \tag{53}$$

where the LHS can be computed from microcanocical MD and allows to determine the coefficients $\gamma_{||}$ and $\gamma_{\perp}$.

We compute the integral (51) as a function of time for the blob-force-force autocorrelation functions (FFAs) and blob-force-force crosscorrelation functions (FFCs), respectively, for $N_{\mathrm{cg}}=24$ and $N_{\mathrm{cg}}=96$. The approximation (51) is identical to the Onsager expression (52) as a function of time. We have computed the Onsager expression both as a numerical derivative of the VV by using a forward difference and also directly as the blob-velocity-force correlation function (VF) $\langle\mathbf{V}_{\mu}\mathbf{F}_{\nu}(\tau)\rangle$. As expected, both results and the Green–Kubo integral (51) are identical within the numerical error. The resulting VFs are plotted in Figure 2 and 3.

In order to determine the friction matrix from (51) or equivalently from (52), we have to choose the right value for τ. By examining the functions in Figure 2 and 3 obtained for the graphene blobs with $N_{\mathrm{cg}}=24$ and $N_{\mathrm{cg}}=96$ this turns out to be difficult due to the highly oscillatory character of the correlations. This is a known phenomenon for covalently bonded atomic lattices. It can be traced back to the optical high frequency modes of the lattice.[15,32,35] In simple fluids without significant elastic properties this effect is expected to be absent. For polymers, depending on their topology, a behaviour between fluids and atomic lattices should be expected for the VF.

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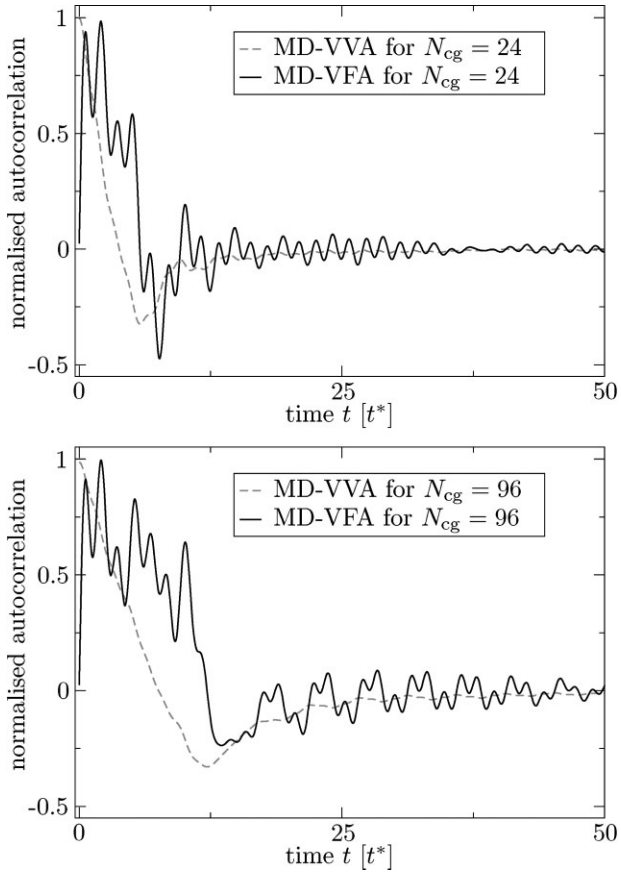


Figure 2. Blob-velocity-force autocorrelation functions (VFAs) for $N_{cg} = 24$ and $N_{cg} = 96$, together with the corresponding blob-velocity-velocity autocorrelations functions (VVAs). The VFAs represent the result of Equation (51 and 52) in dependence of time, for the case $\mu = \nu$. The VVAs have been normalized to $2k_B\,T/M$, while the VFAs have been scaled to ideally fit into the plot range.

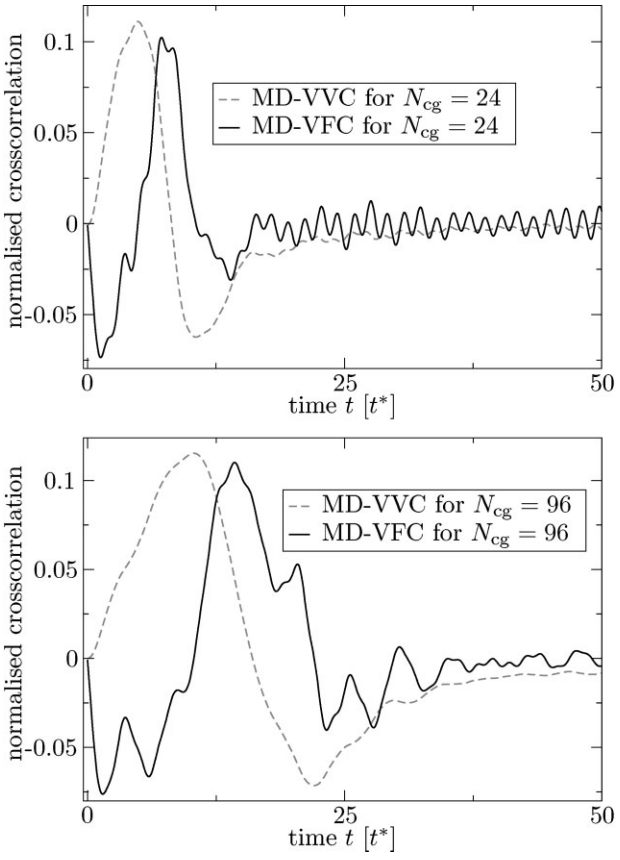


Figure 3. Blob-velocity-force crosscorrelation functions (VFCs) for $N_{cg} = 24$ and $N_{cg} = 96$, together with the corresponding blob-velocity-velocity cross-correlation functions (VVCs). The VFCs represent the result of Equation (51 and 52) in dependence of time, for the case that μ and ν represent neighbouring blobs. The VVCs have been normalized to $2k_B\,T/M$, while the VFCs have been scaled to ideally fit into the plot range.

Ignoring the oscillations on the fast timescales, we expect a plateau region for the VF at times close to zero, but larger than the correlation time τ_F of the FF. The value at this plateau should give the desired friction matrix. Afterwards this plateau value should decay slowly. Indeed this behaviour can be vaguely observed in Figure 2 and 3 and the force-correlation time is $\tau_F \approx 1.5$. But without any additional averaging or smoothing procedure[15,32,35], unique values for the friction matrix can not be determined.

Smoothing of Optical Oscillations

Despite its purpose to remove the optical oscillations, a smoothing can hardly be physically justified. On the other hand, there is one smoothing procedure with a clear physical meaning, which is to integrate the VF once in time in order to obtain the VV. In signal-processing terms a time-integration is nothing else than the application of a low-pass filter. Hence, for the special case of atomic lattices as the honeycomb lattice used here, applying a linear regression to the VV is more practical for obtaining the friction matrix. Of course one may argue that the linear regression shown below is a smoothing procedure. But at least it is clearly physically motivated by the Onsager and the Einstein–Helfand route, hence compatible with our expectation that the VVs are approximately linear for times that are small but larger than τ_F.

We can further illustrate this argument by plotting the Fourier transforms of the FFs, VFs and VVs. This is shown for the autocorrelations in Figure 4. The cross-correlations (not shown) behave similarly. For the FFs we observe that the number of resonance peaks is increasing for larger N_{cg}. Starting from the high frequencies, the first peak is found approximately at the same frequency $f_0 \approx 0.6$ for blobs of all sizes,[15] including $N_{cg} = 1$, i.e., the single C-atom (not shown). This is the optical mode of oscillation of pairs of C-atoms. The additional maxima in the frequency spectrum are due to the different modes of vibration of the blobs. A

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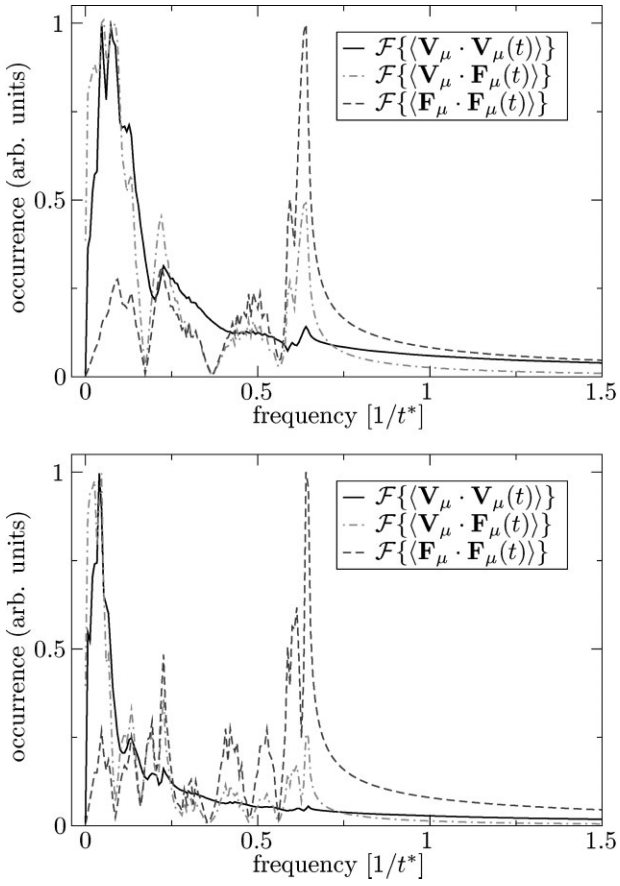


Figure 4. Fourier transforms of the FFAs, VFAs and VVAs for $N_{cg} = 24$ (top) and $N_{cg} = 96$ (bottom). All functions have been scaled in such a way that their maxima are of the same order.

similar behaviour but with already reduced optical peaks is observed for the VFs. This is the effect of one time-integration in the time domain, i.e. one multiplication with $1/i\omega$ in Fourier space. The spectrum of the VV shows the effect of a second time-integration. Basically only the low-frequency modes still have significant amplitudes leading to strongly reduced optical oscillations in the VVs compared to the VFs. The difference in the spectra of the VVs and FFs is also a further indication for the existence of a separation of time scales between these two correlation functions in the time-domain.

Onsager and Einstein–Helfand for Blobs

For the Einstein–Helfand form (38) we get for the graphene blobs

$$\Gamma_{\mu\nu} = \frac{\langle [\mathbf{P}_\mu(\tau_{EH}) - \mathbf{P}_\mu(0)][\mathbf{P}_\nu(\tau_{EH}) - \mathbf{P}_\nu(0)]\rangle}{2k_B T \tau_{EH}} = \frac{[\mathbf{C}^{\mathbf{P}}_{\mu\nu}(\tau_{EH}) - \mathbf{C}^{\mathbf{P}}_{\mu\nu}(0)]}{k_B T \tau_{EH}} \tag{54}$$

because for the relevant part $\Omega^{PP}_{\mu\nu}$ of $\Omega_{\mu\nu}$ we have

$$\Omega^{PP}_{\mu\nu} \equiv < \mathbf{P}_\nu \mathbf{X}_{\nu'} >^{-1'} < \mathbf{X}_{\nu'} iL \mathbf{P}_\mu > + < \mathbf{P}_\nu \mathbf{P}_{\nu'} >^{-1} < \mathbf{P}_{\nu'} iL \mathbf{P}_\mu > = \mathbf{0}. \tag{55}$$

In Equation (54), the time τ_{EH} must be significantly larger than $\tau_{\mathbf{F}}$ but may not reach beyond the time when $\mathbf{C}^{\mathbf{P}}_{\mu\nu}(t) \equiv \langle \mathbf{P}_\mu \mathbf{P}_\nu(t)\rangle$ deviates from being approximately linear. Hence, only if we may choose $\tau_{EH} \gg \tau_{\mathbf{F}}$ the Einstein–Helfand form is equivalent to the Green–Kubo and Onsager form. This is equivalent to saying that there is a separation of timescales between the relevant and irrelevant variables. Otherwise, the error of the Einstein–Helfand approach of ignoring the correlation time $\tau_{\mathbf{F}}$ in which the slope of $\mathbf{C}^{\mathbf{P}}_{\mu\nu}(t)$ increases from zero to the value that Equation (52) predicts, may be significant. Below, we will check this for our system where the separation of timescales exists, but is not very large.[15]

The Einstein–Helfand form (54) is basically a linear regression. The very same technique can also be used as the simplest numerical approximation of the Onsager form (52). The difference is that the Onsager form does not dictate that the linear regression must start at $t = 0$. Hence, we can choose for example to approximate (52) by

$$\Gamma_{\mu\nu} \approx \frac{\left[\mathbf{C}^{\mathbf{P}}_{\mu\nu}(a\tau_{\mathbf{F}}) - \mathbf{C}^{\mathbf{P}}_{\mu\nu}(\tau_{\mathbf{F}})\right]}{k_B T (a-1)\tau_{\mathbf{F}}} \tag{56}$$

where $a > 1$. This takes into account that Equation (52) is only valid for times larger than $\tau_{\mathbf{F}}$.

For $N_{cg} = 24$ and $N_{cg} = 96$ Figure 5 and 6 plot the VVAs and VVCs, respectively, as well as the linear regressions according to Equation (54) and (56) with $\tau_{EH} = a\tau_{\mathbf{F}}$. This means we choose equal upper time bounds for (54) and (56) and compare the difference between the two approaches. Specifically, we choose $\tau^{24}_{\mathbf{F}} = 1.0$ and $a_{24} = 2.5$ for $N_{cg} = 24$ and $\tau^{96}_{\mathbf{F}} = 1.5$ and $a_{96} = 4.3$ for $N_{cg} = 96$. In fact, the force correlation time $\tau_{\mathbf{F}}$ should be independent of N_{cg} and it approximately is $\tau_{\mathbf{F}} \approx 1.5$ for our system.[15] But for $N_{cg} = 24$ it did not harm to lower this value to 1.0 in order to make the linear regression of the Onsager expression (52) more robust. The value of a was estimated in such a way that $\tau_{EH} = a\tau_{\mathbf{F}}$ is right before that time when the VFs from Figure 2 and 3 show their first significant decay, i.e., before the end of the 'plateau region'.

We define the error of the Einstein–Helfand approach with respect to the Onsager approach as the normalized difference $\Delta \equiv (|b_{OS}| - |b_{EH}|)/|b_{OS}|$ where b_{OS} and b_{EH} are the slopes determined from the linear regression of the correlation functions for the Onsager and Einstein–Helfand expressions, respectively. The results are listed in Table 1. Two major trends can be observed: (i) Δ decreases with increasing coarse-graining number N_{cg}. (ii) Even more

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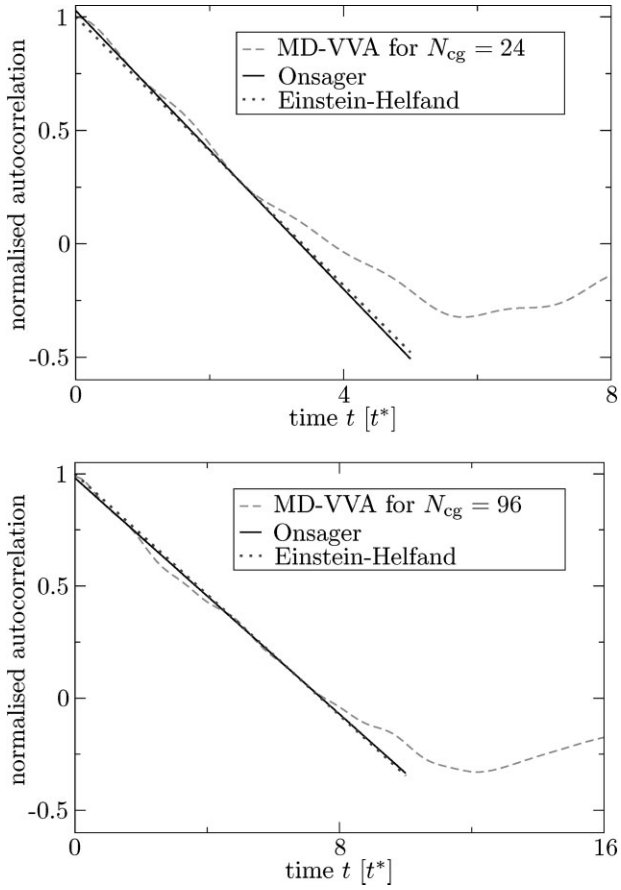


Figure 5. VVAs and linear regression at the origin for $N_{cg} = 24$ and $N_{cg} = 96$. The VVAs have been normalized to $2k_B\,T/M$. Note the different timescales of the plots.

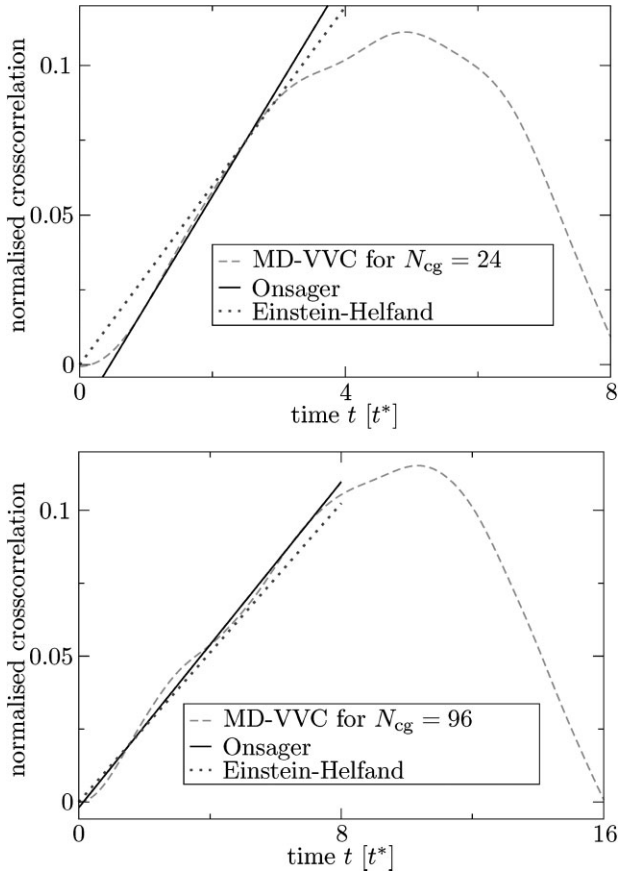


Figure 6. VVCs and linear regression at the origin for $N_{cg} = 24$ and $N_{cg} = 96$. The VVCs have been normalized to $2k_B\,T/M$. Note the different timescales of the plots.

striking is the fact that the error for the VVCs is significantly higher than for the VVAs. Trend (i) tells us that the separation of timescales between the variables treated as relevant and those treated as irrelevant increases with N_{cg} making a Markovian approximation more and more reliable. This is in line with previous findings.[15] Observation (ii) is a further indication that the blob-variables have difficulties in reproducing correctly the sound-propagation in the material, as already observed in previous work.[15]

Results

Based on the relative entropy minimization method proposed by Shell[14] we have optimized a parametric model for the conservative force in ref.[15]. The conservative force between the blobs derives from the model potential

$$V^{cg}(R) = \frac{k_s^{cg}}{2}\sum_{P_{\mu\nu}}(R_{\mu\nu} - r_0^{cg})^2 + \frac{k_\alpha^{cg}}{2}\sum_{T_{\mu\nu\lambda}}(\cos\alpha_{\mu\nu\lambda} - \cos\alpha_0^{cg})^2 \quad (57)$$

with the two parameters k_s^{cg} and k_α^{cg}. The equilibrium distance between blobs is $r_0^{cg} = \sqrt{6N_{cg}}\,r_0/2$, and the equilibrium angle between neighbouring blobs is $\alpha_0^{cg} = \pi/3$. By Shell's entropy minimization and a Newton–Raphson method our optimization resulted in $k_s^{cg} = 0.78, k_\alpha^{cg} = 10.0$ for $N_{cg} = 24$ and $k_s^{cg} = 1.13, k_\alpha^{cg} = 25.5$ for $N_{cg} = 96$.

Together with this model for the conservative force, the computation of the friction matrix according to (56) allows us to construct the DPD-model (44). In order to test how well the DPD model reproduces the MD results for the

Table 1. Difference between the slopes of the correlation functions, as computed from the Onsager and the Einstein–Helfand expression for the friction tensor.

N_{cg}	Function	Δ [%]
24	VVA	3.6
24	VVC	18.4
96	VVA	−2.2
96	VVC	8.3

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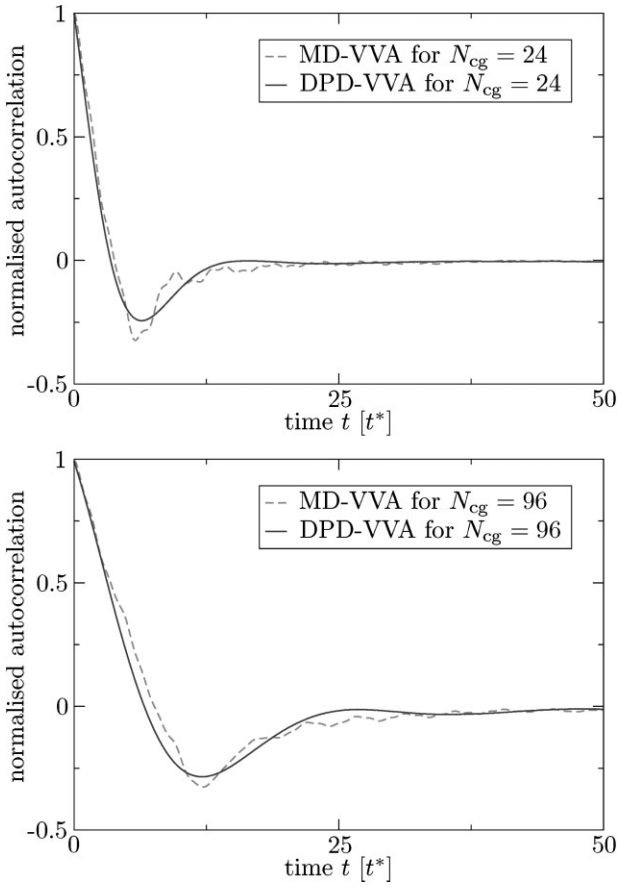


Figure 7. Velocity auto-correlation functions of blobs (VVAs) for $N_{cg} = 24$ (top) and $N_{cg} = 96$ (bottom), respectively, normalized with $2k_B T/M$. MD-VVA stands for the VVA as computed from MD. DPD-VVA is the VVA as computed from DPD.

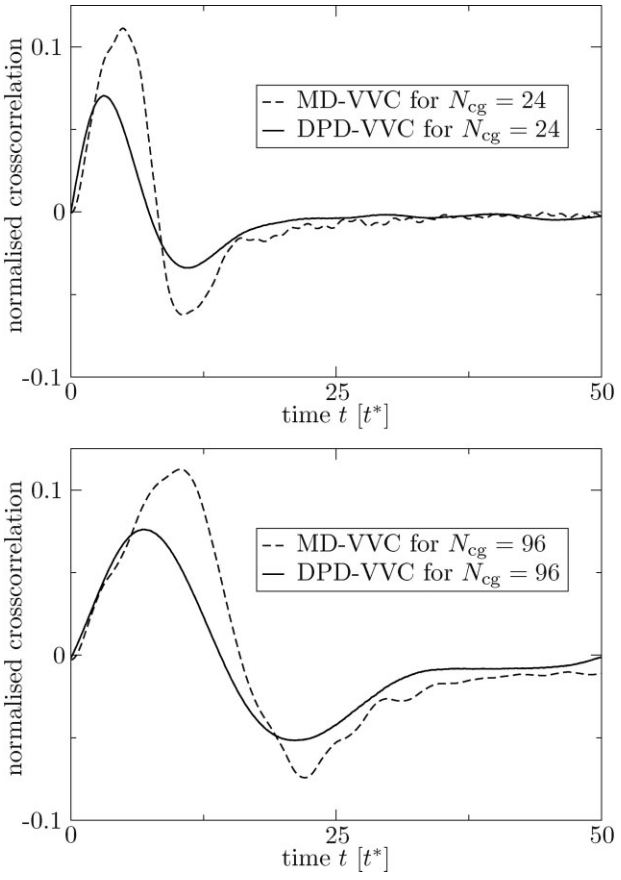


Figure 8. Velocity cross-correlation functions $\langle \mathbf{V}_\mu \mathbf{e}^o_{\mu\nu} \mathbf{V}_\nu(t) \mathbf{e}^o_{\mu\nu} \rangle$ of neighbouring blobs (VVCs) for $N_{cg} = 24$ (top) and $N_{cg} = 96$ (bottom), respectively, normalized with $2k_B T/M$. MD-VVC stands for the VVC, as computed from MD. DPD-VVC is the VVC, as computed from DPD.

dynamics of the blobs we compare the VVAs and VVCs computed from DPD and MD. This is presented in Figure 7 and 8. Here, we briefly summarize the main results, which are discussed in more detail in ref.[15]

A very good agreement between the MD and DPD results is obtained for the VVAs in Figure 7, with the DPD model reproducing the correct description of the observed anti-correlation effect in the blob velocity. It should be clear that the input information from MD is obtained from the decay at $t \approx 0$. The correct description of the auto-correlation at later times is a *prediction* of the DPD model.

The reproduction of the cross-correlations in Figure 8 is qualitatively correct, but exhibits a quantitatively higher dissipation, which also shows up in the decay of shear and pressure waves. On the other hand, it can also be shown that a CG model without dissipative and noise terms shows totally different results than MD.[15] This indicates that dissipation and thermal noise play an important role for the dynamics of the blobs.

Conclusion

Coarse-grained dynamics of molecular systems requires the computation of the friction matrix. We have shown how this matrix arises from projection operator theory when using Mori's or Zwanzig's projector, and how these two projectors are related. Two approximations are required to get the Mori SDE from Zwanzig's SDE. The first assumption is that constrained averages can be approximated by equilibrium averages. The second assumption is that the entropy is a quadratic function of the relevant variables. Both assumptions are expected to be valid near equilibrium when thermal fluctuations are small.

From a practical point of view the question is whether we should use the Zwanzig or Mori approach. Zwanzig's SDE is certainly more fundamental than Mori's SDE and captures non-linear effects and far from equilibrium situations. In Mori theory, only a 'small oscillations' approximation is obtained for the effective potential. In the Zwanzig SDE, the friction matrix depends, in general, on the whole set of CG

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variables. In the Mori SDE, the friction matrix is independent of the CG variables. Hence, Zwanzig's approach is more general. On the other hand, this generality does not come for free: the state dependence of the effective potential and the friction matrix represent a serious practical drawback of Zwanzig's theory. For a large number of CG variables, such as in the presented example of a CG system of graphene blobs, sampling of the CG state space, in order to obtain the functional form of the effective potential and friction, is computationally unfeasible. Two ways out of this dimensionality problem can be envisaged. The first is to formulate approximate parameterized models. In our example, this approach was applied to the computation of the effective potential, using Shell's method[14] and was presented in ref.[15] For the friction matrix, we used the second way, which is to restrict ourselves to cases with small fluctuations of the CG variables. We are then allowed to approximate Zwanzig's expression for the friction matrix with Mori's state-independent expression.

In the Mori picture, we have presented three routes for the calculation of the friction matrix and while they seem to operate in different scales, it is shown that the three routes are entirely equivalent. However, in practice one route may be more convenient than others. For instance in our graphene example where the frictional and stochastic forces are of the DPD-form, applying a linear regression of the velocity–velocity correlation function (VV) near $t = 0$ is more practical than computing the time integral of the force-force correlation (Kirkwood-approximation of the Green–Kubo integral) or to compute the velocity-force correlation for small times (Onsager expression).

The reason are the optical modes which survive even when examining time-correlations of CG variables. This is an effect rather specific for atomic lattices. In fluids, a different behaviour is expected. The rule of thumb is that optical oscillations are dominant as long as the time-correlations directly contain microscopic contributions. This is the case for any correlation function containing the microscopic total force on the blobs $\mathbf{F}_\mu$, hence for the correlation functions $\langle \mathbf{F}_\mu \mathbf{F}_\nu(t) \rangle$ and $\langle \mathbf{V}_\mu \mathbf{F}_\nu(t) \rangle$ required, respectively, for the Kirkwood and the Onsager expression. Hence the advice is to analyse correlation functions containing only slow variables, such as the VV. In the language of signal processing, one could think of the ideal way to filter out the optical modes. A physical approach is the time integration to obtain the VV, which is clearly preferable to any filtering scheme introducing unphysical parameters.

Our approach turns out to be in fact a numerical approximation of the Onsager expression, clearly inspired by the Einstein-Helfand form. In contrast to the Einstein–Helfand form, the linear regression of the VV is not performed by using its value at $t = 0$. Ideally this should not alter the results we obtain. But in practice, the force correlation time $\tau_{\mathbf{F}}$ after which the Markovian assumption holds is finite. This leads to a parabolic shape of the VV with a slope equal to zero at $t = 0$. As a consequence, the predictions of the original Einstein-Helfand approach and of the linear regression of the Onsager expression differ the more the worse the approximation. We presented this fact for the CG dynamics of graphene blobs containing 24 and 96 carbon atoms. In both cases, the difference is noticeable, and decreases with increasing blob size.

The presented results and the theory also suggest routes for further improvement of the coarse-graining of the graphene lattice. One quite straightforward possibility is to compute the Green–Kubo expression instead of its Kirkwood approximation. This requires to compute the correlation function of the force fluctuations $\delta \mathbf{F}_\mu$. As can be seen from Equation (49) Mori theory requires to replace the non-linear angularly dependent model for the conservative force $\mathbf{F}^C_\mu$ used in ref. [15] by the linear Mori-form $\mathbf{F}^C_\mu = k_B T \mathbf{X}_\nu \langle \mathbf{X}_\nu \mathbf{X}_\mu \rangle^{-1}$. This immediately leads to the further question about the difference between these two choices for the conservative force, and of course, which one is better. Again, Shell's method[14] provides us with a means to answer this question by performing 'measurements' from MD simulations.

Being a genuinely two-dimensional material, graphene lends itself to direct MD simulation more easily than 3D bulk materials. Nevertheless, the simulation of molecular engineering devices would still draw major computational benefits from the resort to CG models since these devices live on length and time scales hardly reachable by sheer MD.[26,27] A coupling to a surrounding medium as, e.g., approached in ref. [36] only adds to the need for CG models.

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